Linear Regression

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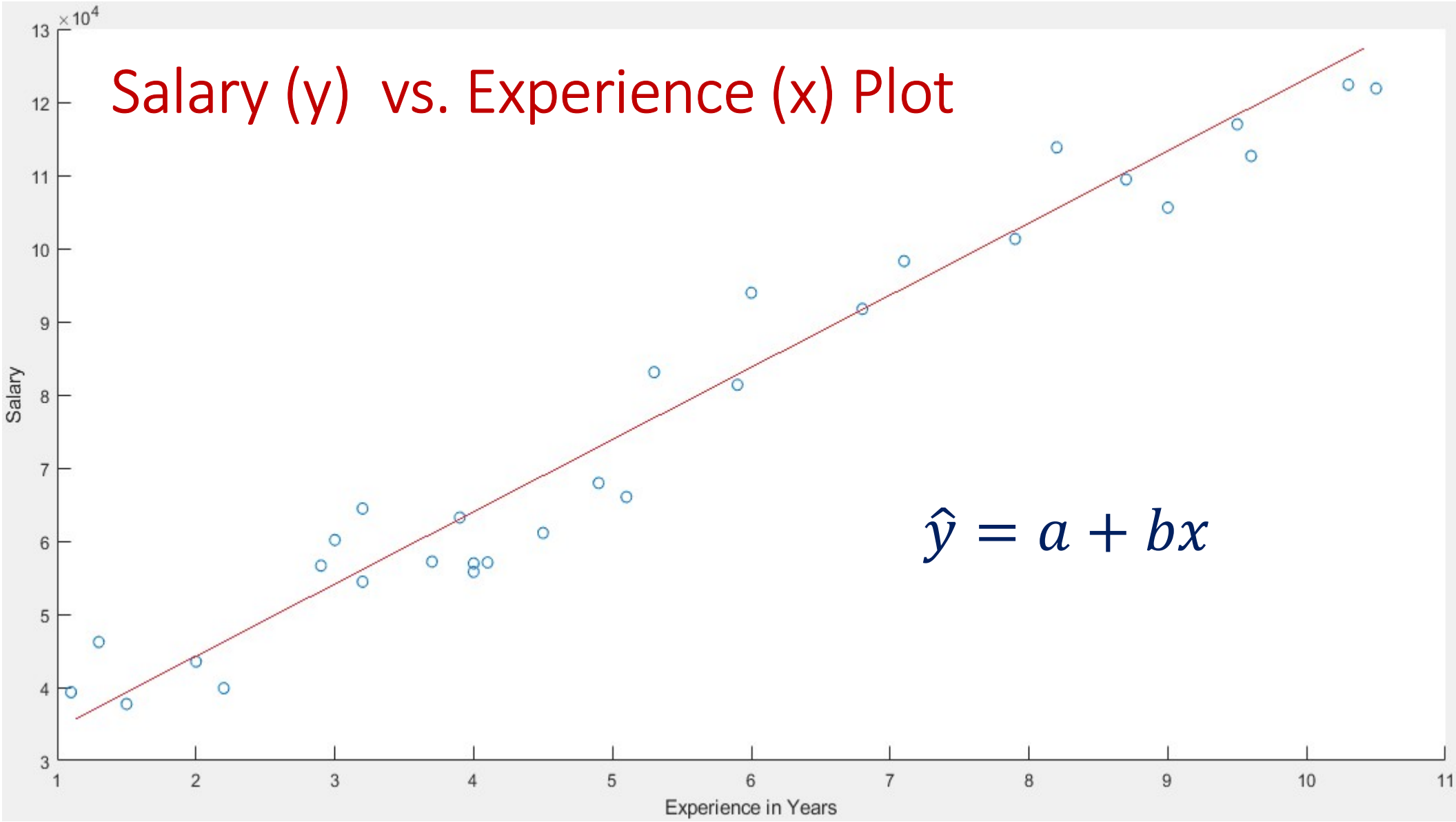
Overview

- The Linear Regression Problem
- Least Squares Problem Formulation
- Partial Differentiation Approach
- Concept of Pseudo Inverse

Salary (y) vs. Experience (x) Dataset

Experience	Salary		Experience	Salary
1.1	39343		4.9	67938
1.3	46205		5.1	66029
1.5	37731		5.3	83088
2	43525		5.9	81363
2.2	39891		6	93940
2.9	56642		6.8	91738
3	60150		7.1	98273
3.2	54445		7.9	101302
3.2	64445		8.2	113812
3.7	57189		8.7	109431
3.9	63218		9	105582
4	55794		9.5	116969
4.1	57081		9.6	112635
4.5	61111		10.3	122391
			10.5	121872

Salary (y) vs. Experience (x) Plot



$$\hat{y} = a + bx$$

Approach #1

Given Dataset

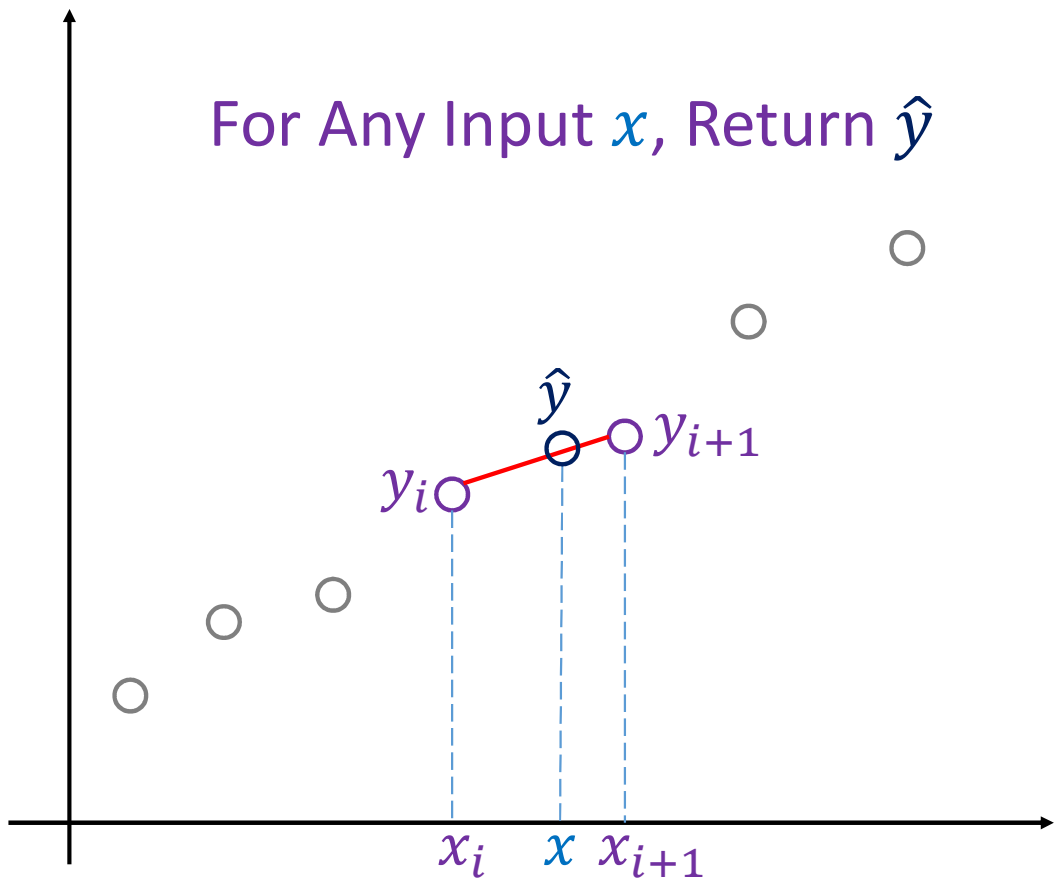
$$(x_i, y_i); i = 1, \cdots n$$

For Any Input x, Return $\hat{y} = \bar{y}$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

Everyone earns 76003 units of salary irrespective of experience !!!

Approach #2

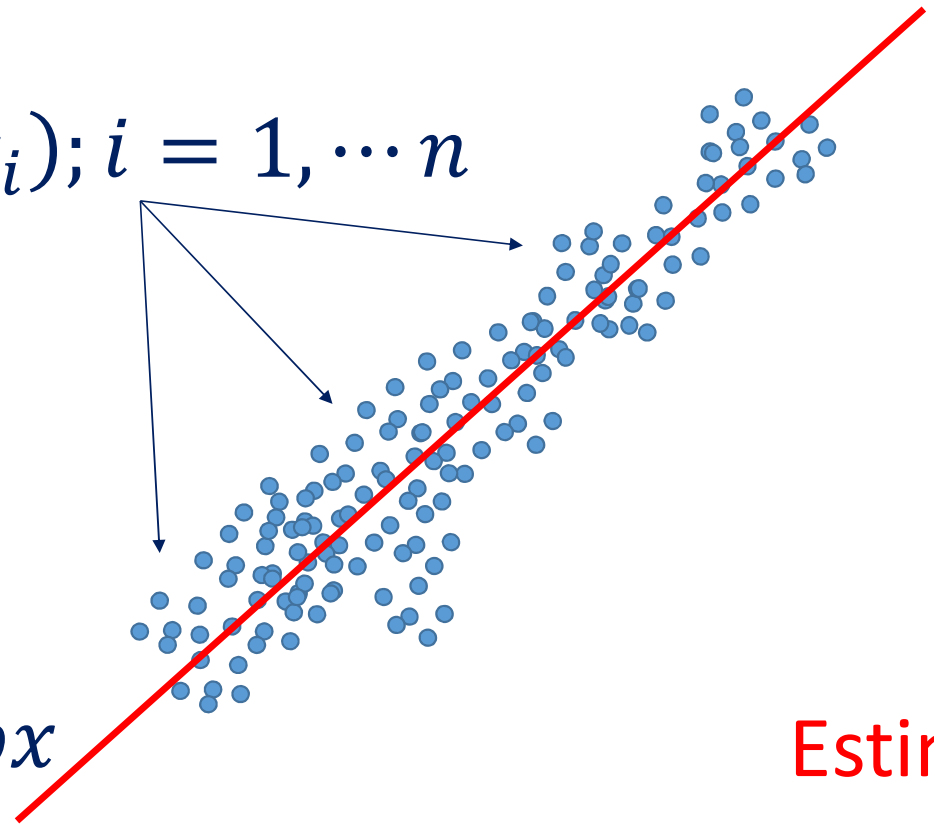


Compute $\hat{y}$ by Linear Interpolation

$$\frac{\hat{y} - y_i}{x - x_i} = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}$$

Linear Regression: Problem Definition

$(x_i, y_i); i = 1, \cdots n$



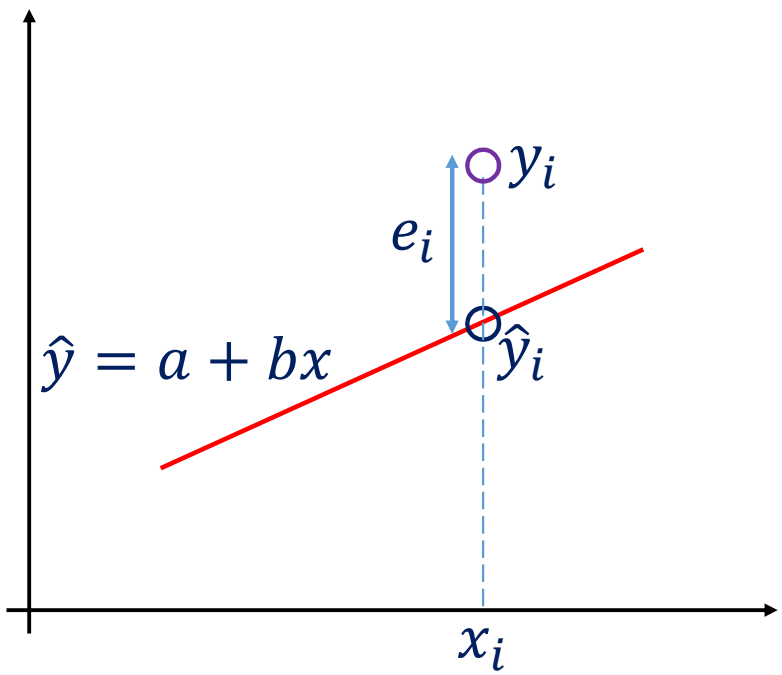
$y = a + bx + \eta$

$\eta \sim \aleph(0, \sigma^2)$

$y = a + bx$

Estimate a & b in presence of noise η

Linear Regression: Problem Formulation



Point Error

$$e_i = y_i - (a + bx_i)$$

Any value of (a,b) will provide us with some point error for given (x_i, y_i)

Total Error

$$E = \frac{1}{n}\sum_{i=1}^{n} e_i^2$$

Minimize E w.r.t (a, b)

Differentiating w.r.t. $\boldsymbol{a}$ and $\boldsymbol{b}$

$$E = \frac{1}{n}\sum_{i=1}^{n} e_i^2 = \frac{1}{n}\sum_{i=1}^{n}(y_i - a - bx_i)^2$$

$$\frac{\partial E}{\partial a} = ?$$

$$\frac{\partial E}{\partial b} = ?$$

Differentiating w.r.t. $\boldsymbol{a}$

$$\frac{\partial E}{\partial a} = \frac{1}{n}\sum_{i=1}^{n} 2e_i \frac{\delta}{\delta a}\{y_i - a - bx_i\} = \frac{2}{n}\sum_{i=1}^{n} e_i\{-1\}$$

$$\Rightarrow \quad \frac{\partial E}{\partial a} = -\frac{2}{n}\sum_{i=1}^{n}\{y_i - a - bx_i\}$$

Setting $\frac{\partial E}{\partial a}$ to Zero $\Rightarrow$ $-\frac{2}{n}\sum_{i=1}^{n}\{y_i - a - bx_i\} = 0$

$$\Rightarrow \quad \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{1}{n}\sum_{i=1}^{n} a + \frac{1}{n}\sum_{i=1}^{n} bx_i \quad \Rightarrow \quad \boxed{\bar{y} = a + b\bar{x}}$$

Differentiating w.r.t. $\boldsymbol{b}$

$$\frac{\partial E}{\partial b} = \frac{1}{n}\sum_{i=1}^{n} 2e_i \frac{\delta}{\delta b}\{y_i - a - bx_i\} = \frac{2}{n}\sum_{i=1}^{n} e_i\{-x_i\}$$

$$\Rightarrow \quad \frac{\partial E}{\partial b} = -\frac{2}{n}\sum_{i=1}^{n}\{y_i - a - bx_i\}x_i$$

Setting $\frac{\partial E}{\partial b}$ to Zero $\Rightarrow$ $-\frac{2}{n}\sum_{i=1}^{n}\{y_i - a - bx_i\}x_i = 0$

$$\Rightarrow \quad \frac{1}{n}\sum_{i=1}^{n} x_i y_i = \frac{1}{n}\sum_{i=1}^{n} a x_i + \frac{1}{n}\sum_{i=1}^{n} b x_i^2 \quad \Rightarrow \quad \boxed{\overline{xy} = a\bar{x} + b\overline{x^2}}$$

Final Solution

$$\bar{y} = a + b\bar{x}$$

$$\overline{xy} = a\bar{x} + b\overline{x^2}$$

Simultaneous Equations

$$b = \frac{\overline{xy} - (\bar{x})(\bar{y})}{\overline{x^2} - (\bar{x})^2}$$

$$a = \bar{y} - b\bar{x}$$

$$\begin{bmatrix} 1 & \bar{x} \\ \bar{x} & \overline{x^2} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} \bar{y} \\ \overline{xy} \end{bmatrix} \Rightarrow \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix}$$

Numerical Example: Dataset

X	-1.313	8.585	5.353	10.00	-8.383
Y	-1.062	39.506	27.089	46.69	-28.019

Linear Model: $Y = a + bX$

Dataset Size: $n = 5$

Numerical Example: Computations

$$\begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix} = \begin{bmatrix} 5 & 14.2420 \\ 14.2420 & 274.3555 \end{bmatrix}$$

$$\begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix} = \begin{bmatrix} 84.2040 \\ 1187.3441 \end{bmatrix}$$

Numerical Example: Solution

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 & 14.2420 \\ 14.2420 & 274.3555 \end{bmatrix}^{-1} \begin{bmatrix} 84.2040 \\ 1187.3441 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5.2968 \\ 4.0528 \end{bmatrix}$$

$$\boxed{Y = 5.2968 + 4.0528X}$$

Numerical Example: Evaluation

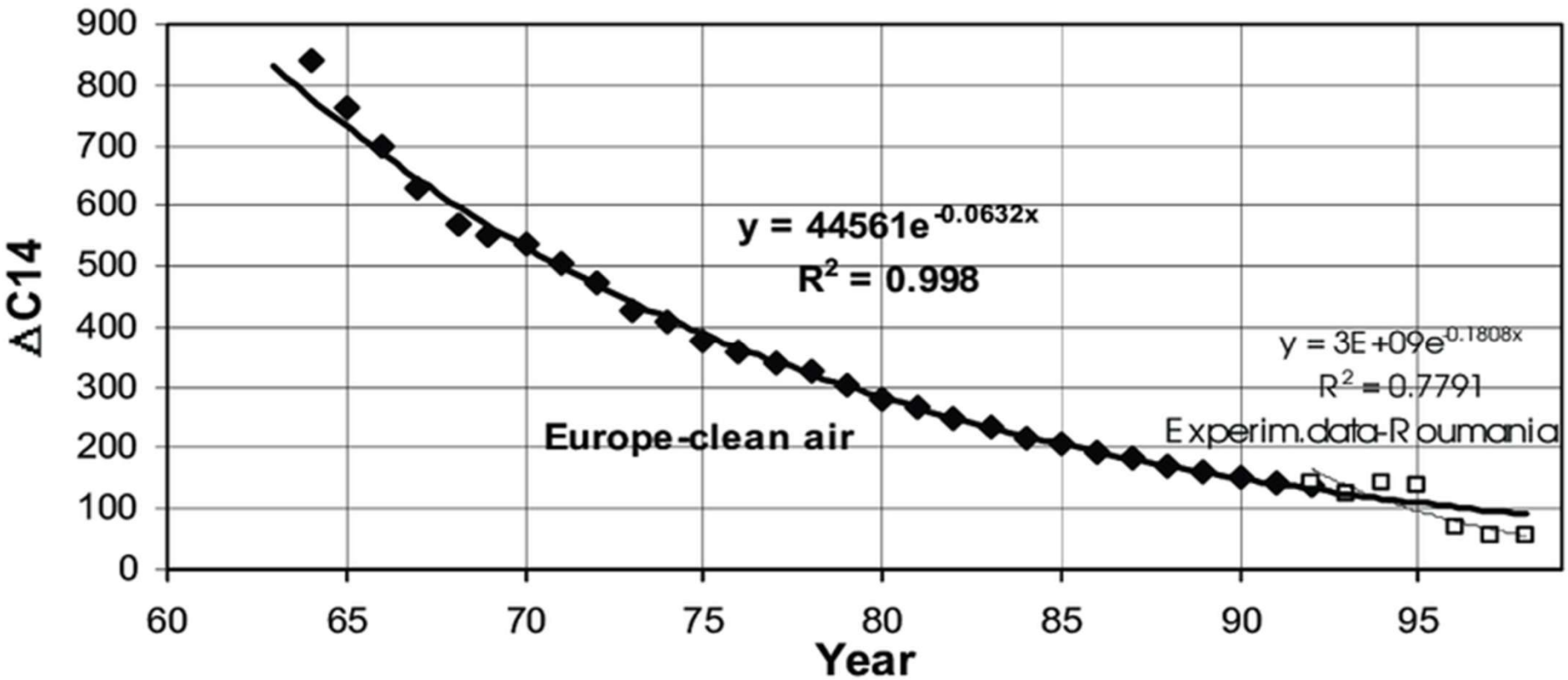
X	-1.313	8.585	5.353	10.00	-8.383
Y	-1.062	39.506	27.089	46.69	-28.019
$\hat{Y}$	-0.0245	40.0901	26.9914	45.8248	-28.6778
E	-1.0375	-0.5841	0.0976	0.8652	0.6588

Learned Linear Model: $Y = 5.2968 + 4.0528X$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{2.6096}{3842.6235} = 0.9993$$

Can be Solved by Linear Regression?

$S = \{(v_i, t_i); i = 1, \dots n\}$ Model: $v = ae^{bt}$



Can be Solved by Linear Regression?

$S = \{(v_i, t_i); i = 1, \dots n\}$ Model: $v = ae^{bt}$

$$\log(v) = \log(a) + bt$$

$$\log(v) = \log(a) + bt$$

$$\hat{z} = A + bt$$

House Price Dataset

$$Y = a_0 + a_1 * (BEDROOM) + a_2 * (ROOM) + a_3 * (SPACE) + a_4 * (TAX) + a_5 * (GARAGE) + a_6 * (CONDITION)$$

Single Dependent Variable,
Multiple Independent Variables

Linear Regression: Extending the Problem

- Consider linear regression problems involving m independent variables

$$y = a + \sum_{k=1}^{m} b_k x_k$$

- For large m, solving $(m+1)$ different equations (obtained by partial differentiation) is a tedious approach
- This necessitates a compact representation leading to a well defined solution
- The approach should also be realizable on a programmable digital device

The Linear Regression Problem

$$y = a + \sum_{k=1}^{m} b_k x_k$$

Linear Regression Problems involving m Independent Variables

$$\left\{\left(x_1^{(i)}, \dots x_m^{(i)}, y^{(i)}\right); i = 1, \dots n\right\}$$

The dataset containing n training instances

$$e_i = y^{(i)} - \left\{a + \left(b_1 x_1^{(i)} + \cdots + b_m\, x_m^{(i)}\right)\right\}$$

Linear Model Prediction Error for Instance i

Linear Regression: Reformulation of Error

$$E = \begin{bmatrix} e_1 \\ \vdots \\ e_i \\ \vdots \\ e_n \end{bmatrix} = \underbrace{\begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(i)} \\ \vdots \\ y^{(n)} \end{bmatrix}}_{\substack{\mathbf{Y} \\ n \times 1}} - \underbrace{\begin{bmatrix} 1 & x_1^{(1)} & \cdots & x_m^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_1^{(i)} & \cdots & x_m^{(i)} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_1^{(n)} & \cdots & x_m^{(n)} \end{bmatrix}}_{\substack{\mathbf{A} \\ n \times (m+1)}} \underbrace{\begin{bmatrix} a \\ b_1 \\ \vdots \\ b_m \end{bmatrix}}_{\substack{\mathbf{P} \\ (m+1) \times 1}}$$

Linear Regression: Reformulation of Error

$$E = \begin{bmatrix} e_1 \\ \vdots \\ e_i \\ \vdots \\ e_n \end{bmatrix} = \begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(i)} \\ \vdots \\ y^{(n)} \end{bmatrix} - \begin{bmatrix} 1 & x_1^{(1)} & \cdots & x_m^{(1)} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_1^{(i)} & \cdots & x_m^{(i)} \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_1^{(n)} & \cdots & x_m^{(n)} \end{bmatrix} \begin{bmatrix} a \\ b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$e_i = y^{(i)} - \left\{ a + \left(b_1 x_1^{(i)} + \cdots + b_m x_m^{(i)} \right) \right\}$$

Linear Regression: Reformulation of Error

$$\mathrm{E_t} = E^T E = \sum_{i=1}^{n} e_i^2$$

The Total Error

Minimize $\mathrm{E_t} = E^T E$ w.r.t. P

Minimization of Total Error

$$E_t = E^T E = (Y - AP)^T (Y - AP)$$

$$\Downarrow$$

$$E_t = ?$$

Minimization of Total Error

$$E_t = E^T E = (Y - AP)^T (Y - AP)$$

$$\Downarrow$$

$$E_t = (Y^T - P^T A^T)(Y - AP)$$

$$\Downarrow$$

$$E_t = Y^T Y - Y^T AP - P^T A^T Y + P^T A^T AP$$

Minimization of Total Error

$$E_t = Y^T Y - Y^T AP - P^T A^T Y + P^T A^T AP$$

$$\Downarrow$$

$$\nabla_P E_t = ?$$

Minimization of Total Error

$$E_t = Y^T Y - Y^T AP - P^T A^T Y + P^T A^T AP$$

$$\Downarrow$$

$$\nabla_P(E_t) = -A^T Y - A^T Y + \{A^T A + (A^T A)^T\}P$$

$$\Downarrow$$

$$\nabla_P(E_t) = -2A^T Y + 2(A^T A)P$$

Minimization of Total Error: Pseudo-Inverse

$$\nabla_P(E_t) = -2A^TY + 2(A^TA)P = \mathbf{0}$$

Setting $\nabla_P(E_t)$ to Null Vector $\Rightarrow$ $A^TAP = A^TY$

$$P = (A^TA)^{-1}A^TY = A^{\#}Y$$

Pseudo-Inverse of A: $A^{\#} = (A^TA)^{-1}A^T$

Problem Formulation using Pseudo-Inverse

$$S = \{(z_i, x_i, y_i, u_i); i = 1, \dots n\}$$

Model

$$\hat{z} = a + bx + cy + du$$

$$P = (A^T A)^{-1} A^T Y = A^{\#} Y$$

$A = ?$ $\quad$ $Y = ?$ $\quad$ $P = ?$

Problem Formulation using Pseudo-Inverse

$$A = \begin{bmatrix} 1 & x_1 & y_1 & u_1 \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_i & y_i & u_i \\ \vdots & \vdots & \cdots & \vdots \\ 1 & x_n & y_n & u_n \end{bmatrix} \qquad P = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} \qquad Y = \begin{bmatrix} z_1 \\ \vdots \\ z_i \\ \vdots \\ z_n \end{bmatrix}$$

Problem Formulation using Pseudo-Inverse

$$S = \{(y_i, x_i); i = 1, \dots n\}$$

Model

$$\hat{y} = a + bx$$

$$P = (A^T A)^{-1} A^T Y = A^{\#} Y$$

$A = ?$ $\quad$ $Y = ?$ $\quad$ $P = ?$

Final Solution: Pseudo-Inverse for $\boldsymbol{Y} = \boldsymbol{a} + \boldsymbol{bX}$

$$A^T A = \begin{bmatrix} 1 & \cdots & 1 & \cdots & 1 \\ x_1 & \cdots & x_i & \cdots & x_n \end{bmatrix} \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_i \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix}$$

$$A^T Y = \begin{bmatrix} 1 & \cdots & 1 & \cdots & 1 \\ x_1 & \cdots & x_i & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_i \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix}$$

Final Solution: Pseudo-Inverse

$$P = (A^T A)^{-1} A^T Y = A^{\#} Y$$

$$\overset{P}{\begin{bmatrix} a \\ b \end{bmatrix}} = \underbrace{\begin{bmatrix} n & \sum_{i=1}^{n} x_i \\ \sum_{i=1}^{n} x_i & \sum_{i=1}^{n} x_i^2 \end{bmatrix}^{-1}}_{(A^T A)^{-1}} \underbrace{\begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_i y_i \end{bmatrix}}_{A^T Y}$$

Assignments

The linear regression problem involving m independent variables and single dependent variable has the prediction model $\hat{y} = a + \sum_{k=1}^{m} b_k x_k$. Consider a training dataset containing n instances. The solution is obtained by using the pseudo-inverse based formulation $P = (A^T A)^{-1}(A^T Y)$ (notations having their usual meanings as shown in the slides). Evaluate the number of Multiplications and Additions involved in computing P.

Consider the linear regression problem involving single independent and single dependent variable. Prove that the optimal model parameters obtained in Least Squares framework is an unique solution. Can this be extended to linear regression problems involving single dependent and multiple independent variables? Justify with Mathematical proof.

Summary

- The Linear Regression Problem
- Single Dependent Variable, Single Independent Variable
- Least Squares Problem Formulation
- Partial Differentiation Approach
- Single Dependent Variable, Multiple Independent Variables
- Concept of Pseudo Inverse

Thank You